

The Edge Negotiator

Coordinated On-Device SLMs for Multi-Junction Emergency Response

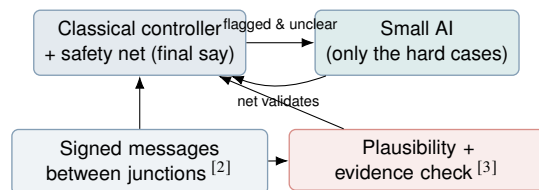
Progress report | MSc Systems Engineering for IoT, UCL (25153651) | Supervisors: Dr A. Delibasi (UCL), L. Stott (Microsoft) | 2026-06-21

1. The project and its goal

The system is a corridor of traffic junctions on one arterial. Its purpose is **coordinated emergency response**: when an ambulance crosses the corridor, or an incident blocks a lane, the junctions share what they see and clear a path together, instead of each reacting alone. Each junction runs a fast classical controller (*MaxPressure*^[1]) by default, which is also the safety floor; a small AI model (**Phi-4-mini**, on the junction via **Microsoft Foundry Local**, never retrained) is called in only for the hard, ambiguous cases a fixed rule cannot settle.

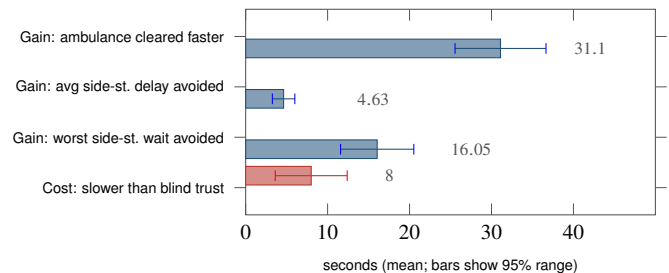
Coordinating means acting on a neighbour's report, which is exactly what a compromised neighbour can abuse. So the enabling property, what makes the coordination safe to deploy, is **robust degradation**: if a junction is hijacked or its messages are tampered with, the system detects that the reports do not match the real traffic and falls back to local control. A deterministic safety net makes every clear-cut call (clear an ambulance a junction physically detects; refuse a claim it cannot verify); safety never depends on the AI. The central question we measure is whether the AI improves the genuinely ambiguous cases over a well-tuned fixed rule, and we report it either way.

2. How it works



The classical controller runs by default and has the final say on every safety call. Only a flagged, unclear alert goes to the AI. Every message is signed and checked first.

3. Measured effects (30 runs each)



SUMO^[4], 30 paired runs. Top: coordinated emergency preemption clears the ambulance faster than the same controller with preemption off. Below: the coordination stays safe under a spoofing neighbour, at a small honest cost.

Every 95% range excludes zero.

4. What we have built, and what it shows

- **Coordinated emergency handling (the core)**: junctions share what they sense; a junction clears an ambulance it physically detects, and a downstream junction pre-clears for it once an upstream junction has independently confirmed the same vehicle.
- **Working and tested**: the classical controller and its safety net, signed messaging between junctions, an approved-senders registry, a check that flags neighbour reports inconsistent with the real traffic, and a four-junction simulated corridor.
- **Live demo**: a real ambulance is cleared across the junctions; a hijacked junction's signed-but-fake "ambulance approaching" alert is refused, so a compromised

neighbour cannot hijack the coordination.

- **Evidence (30 runs)**: coordinated emergency preemption (6 local plus 7 corroborated-downstream clearances in one run) clears the ambulance 31 s faster than with preemption off; under a spoofing neighbour the worst side-street wait is held 16 s below a trust-everything controller; requiring physical evidence costs 8 s of ambulance time (accepted).
- **Independent check**: an automated audit of the write-up found no fabricated sources; a few over-stated citations were corrected. Every result is reproducible from its settings.

5. Where it sits in the literature

To our knowledge, no prior system coordinates several junctions for emergency response using a frozen on-device SLM, and keeps that coordination trustworthy when an approved junction is compromised. The parts exist separately; this combination does not.

Work (year)	What it does	Coord?	Trust?	How we differ
CoLLMLight (2025)	Cooperative LLM junctions share state; fine-tuned 8B	Yes	No	Frozen 3.8B; signed + plausibility-checked channel
VLMLight (2025)	Fast-classical + slow-LLM, safety-critical cases	1 junc.	No	Multi-junction; small on-device model; compromised-channel test
LA-Light (2024)	LLM calls rule/RL for rare events; cloud GPT-4	Multi	No	On-device; faults are adversarial and signed, not benign
iLLM-TSC (2024)	LLM refines an RL controller under noisy comms	No	Noise	Malicious (spoofed) messages, not random noise
LLMLight (2023/25)	Single LLM controls one junction; fine-tuned	No	No	Multi-junction coordination + trust layer
EMVLight (2022)	Trained MARL for emergency-vehicle preemption	Yes	No	No retraining; handles novel incidents; trust layer
REG-TSC (2025)	RAG + fine-tuned 8B agents, emergencies	Yes	No	Frozen 3.8B; RAG optional; signed + checked
CoLight (2019)	Graph-attention RL, cooperative control	Yes	No	Language reasoning on hard cases + security
MaxPressure (2013)	Throughput-optimal classical control	Local	No	Adopted directly as the default and safety floor
SafeLight (2023)	Trained RL + rule-based safety override	No	No	No training; defends against spoofed inputs
Blockchain-TSC (2019)	Blockchain secures vehicle→signal data	No	Yes	Lightweight signed log; secures signal↔signal
proofmember23 (2023)	Device-identity / membership proof (IoT)	No	Yes	Applied to junctions + a content plausibility check
Derhab (2020)	Flow-conservation check for sensor-net attacks	n/a	Yes	Ported to vehicle conservation across junctions
Keijzer (2021)	Theory-only detection of coordination attacks	Yes	Yes	A measured detection-vs-attacker-strength curve; identity + audit

Why this direction. The contribution is the combination: a frozen on-device SLM coordinating several junctions for emergencies and incidents, kept safe when a junction is compromised. Coordination alone is precededented (CoLLMLight: normal flow, cloud-scale, fine-tuned, no trust); emergency handling alone is precededented (VLMLight, LA-Light, EMVLight); the fusion on a frozen edge model under a compromised channel is ours. Racing the classical controller on ordinary traffic is a dead end (throughput-optimal, microsecond-fast), so raw performance is a secondary, honest axis; the AI earns its place on the hard cases, and robustness to a lying neighbour is what makes leaning on a neighbour's report safe. We use SUMO (it models emergency vehicles and lane blockage), a signed hash-chained log not a blockchain (one trust domain), and a permissioned key registry not decentralised identity.

6. Value and how we measure success

The deployable angle: a safety-critical demonstration of edge AI (Foundry Local, Phi-4, no cloud, no retraining) with verifiable agent identity, the same trust problem as Entra Agent ID, made physical and measurable. Success is judged on:

Indicator	Target / acceptance
Coordinated emergency-vehicle delay	Reduced versus the same controller with preemption off (measured: 31 s faster over 30 runs); incident reallocation to follow
Spoofed-emergency resistance	Every uncorroborated emergency claim refused; no false preemption
No-harm guarantee	The security layer does not slow normal traffic beyond a pre-registered margin
Attack-detection quality	Precision / recall / detection-latency reported across a sweep of lie sizes
Edge feasibility	Phi-4-mini decision within budget on Foundry Local; gated to keep a corridor within it
Auditability	Tamper-evident log of every signed message and decision

7. Next steps

- Central experiment: on flagged, unclear alerts, does the AI beat a well-tuned fixed rule? Measure and report either way, including a null.
- Make normal-traffic coordination steer decisions, not only verify them, then measure the green-wave benefit.
- Detection limit: grow the lie until detection fails; report where.
- Real Lambeth arterial and a full demand sweep, with confidence ranges.
- Run the real Phi-4-mini on Foundry Local; confirm repeatable decisions.

8. What we do and do not claim

Claim: coordinated emergency preemption across junctions (a real ambulance cleared, a downstream junction pre-cleared on independent corroboration); authenticated messaging between junctions; detection of faked or faulty reports with a measured reliability curve; a deterministic safety floor; a fair attack-and-defence comparison.

Do not claim: new cryptography; that the AI beats the classical controller on ordinary traffic; that the AI yet improves the ambiguous decisions (measured next); that normal-traffic green-wave coordination yet steers decisions (it is verified end to end but not yet causal, the next build step); defence against colluding insiders or a stolen master key; or that simulator results transfer directly to hardware.

References. [1] Varaiya, Max-pressure control, *Transp. Res. C* 36 (2013). [2] Bernstein et al., Ed25519, *J. Cryptogr. Eng.* (2012). [3] Lighthill, Whitham & Richards (LWR vehicle-conservation model, 1955, 1956). [4] Lopez et al., SUMO, *IEEE ITSC* (2018). [5] Yuan et al., CoLLMLight, arXiv:2503.11739 (2025). [6] Wang et al., VLMLight, arXiv:2505.19486 (2025). [7] Wang et al., LA-Light, arXiv:2403.08337 (2024). Further: iLLM-TSC 2407.06025; LLMLight 2312.16044; EMVLight 2206.13441; REG-TSC 2510.26242; CoLight 1905.05717; SafeLight 2211.10871; Blockchain-TSC 1906.02628; proofmember 2310.08163; Derhab *Sensors* 20(21):6106 (2020); Keijzer *ECC* (2021).